

The Naylor Model

Variable Glossary

Statcast-style reference · v6

Every variable in this glossary tells you:

- what it actually measures (plain English)
 - the units and a typical value
 - what's elite and what's poor
 - a real player example
 - why it affects steal success

No statistical jargon.

Replace 'coef_z' and 'OR/SD' with
'SB % Boost per Tier' and 'Odds Multiplier'.

Speed & Acceleration

Sprint Speed

What it is

Top running speed measured during the player's fastest sprints.

Units	League avg	Elite
feet per second (ft/s)	27.0	29.0 +

Real example

Bobby Witt Jr.: 30.2 (elite) · Josh Naylor: 24.4 (3rd %ile)

Tier chart

Elite	29.0 +
Above avg	27.5 – 29.0
Average	26.5 – 27.5
Below avg	25.5 – 26.5
Poor	≤ 25.5

Why it matters

Faster runners are obviously harder to throw out, but the model finds it matters LESS than raw acceleration on a 90-ft steal. Once you're above ~28 ft/s the extra speed is wasted because you're past second base before the throw arrives anyway.

Source

Baseball Savant Sprint Speed leaderboard

Speed Capped

What it is

Sprint speed, but anything over 28 ft/s is clipped to 28.

Units	League avg	Elite
ft/s	27.0	28.0 (capped)

Real example

Witt 30.2 → capped at 28.0 · Naylor 24.4 → unchanged

Tier chart

 ≥ 28 ft/s	treated as 28.0 (no extra reward)
 < 28 ft/s	kept as-is — every fraction matters

Why it matters

The model adds this version because the marginal benefit of more speed vanishes above 28 ft/s. Capping prevents the model from over-rewarding pure speedsters and lets it find smaller advantages like jump and lead.

Source

computed from Sprint Speed

Jump Time (accel_0_30)

What it is

How fast you cover the first 30 feet from contact.

Units	League avg	Elite
seconds	1.78	≤ 1.65

Real example

Corbin Carroll: 1.67 · Josh Naylor: 1.87 (slow) · Nasim Nuñez: 1.64 (elite)

Tier chart

 Elite	≤ 1.68
 Above avg	1.69 – 1.74
 Average	1.75 – 1.80
 Below avg	1.81 – 1.86
 Poor	≥ 1.87

Why it matters

The single most important running metric for steals — the throw is in the air before you hit top speed, so what happens in the first 30 ft decides everything. In the v5/v6 simple GLM this feature has the strongest weight of any speed variable.

Source

Statcast Running Splits · seconds_since_hit_030

Acceleration Phase (accel_5_30)

What it is

Time from 5 ft to 30 ft — pure first-burst, excluding bat-flip.

Units	League avg	Elite
seconds	1.32	≤ 1.22

Real example

Witt: 1.23 (elite) · Naylor: 1.38

Tier chart

Elite	≤ 1.22
Above avg	1.23 - 1.28
Average	1.29 - 1.34
Below avg	1.35 - 1.40
Poor	≥ 1.41

Why it matters

Cleaner version of Jump Time — removes how quickly the runner cleared the batter's box. Less noisy proxy of biomechanical first-burst.

Source

Statcast Running Splits · 030 - 005

Top-Speed Phase (maintain_30_90)

What it is

Time from 30 ft to 90 ft — back half of the sprint.

Units	League avg	Elite
seconds	2.27	≤ 2.15

Real example

Witt: 2.10 (elite) · Naylor: 2.44

Tier chart

 Elite	≤ 2.15
 Above avg	2.16 - 2.22
 Average	2.23 - 2.30
 Below avg	2.31 - 2.38
 Poor	≥ 2.39

Why it matters

Matters less for SBs than Jump Time — a steal of 2nd only requires covering ~85 ft. But it captures conditioning and shows up in pre-2023 data as a stronger predictor than today.

Source

Statcast Running Splits · 090 - 030

Total 90 (total_90)

What it is

Full home-to-first time from contact.

Units	League avg	Elite
seconds	4.05	≤ 3.95

Real example

Carroll: 3.94 (elite) · Naylor: 4.31 (15th percentile)

Tier chart

Elite	≤ 3.95
Above avg	3.96 - 4.02
Average	4.03 - 4.10
Below avg	4.11 - 4.18
Poor	≥ 4.19

Why it matters

A summary number — equals Jump Time + Top-Speed Phase. Useful for ranking 'overall fast' runners. Naylor's 4.31 puts him in the bottom 25%, yet his accel_gap (below) shows he's still quick OFF THE LINE — the steal-relevant part.

Source

Statcast Running Splits · seconds_since_hit_090

Accel Gap

What it is

How much faster the runner is off the line than their top speed implies.

Units	League avg	Elite
percentile points	0	+ 25 or more

Real example

Naylor: +22 to +30 across seasons · Witt: -20 (slower off the line for his speed)

Tier chart

Naylor archetype	+15 or more (fast jump despite low speed)
Above expected	+5 to +15
As expected	-5 to +5
Below expected	-15 to -5
Pure speedster	≤ -15 (fast top speed but slow start)

Why it matters

This is the heart of the Naylor archetype. POSITIVE values mean the runner is faster off the line than their top speed would predict — exactly what a slow-but-effective stealer looks like. Pure speedsters often have NEGATIVE accel_gap because they coast on top-end speed.

Source

computed: `pctile(Jump Time, inverted) - pctile(Sprint Speed)`

Bolts

What it is

Number of plays in a season where the runner topped 30 ft/s.

Units count	League avg 13	Elite 100 +
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Real example

Witt: 250+ · Carroll: 200+ · Naylor: 0 (never hit 30 ft/s)

Tier chart

 Elite	100 +
 Above avg	30 - 100
 Average	5 - 30
 Below avg	1 - 5
 Never	0

Why it matters

A simple count of elite-speed moments. Big leaderboard presence; the model treats it as a speed credential. Zero bolts doesn't disqualify you (Naylor proves that), but it puts you in the slow-steal archetype the model is designed to identify.

Source

Baseball Savant

Lead Distance (real Baseball Savant)

Primary Lead

What it is

Average distance off first base when the pitcher starts his motion.

Units	League avg	Elite
feet (ft)	11.5	13.0 +

Real example

Trea Turner: 13.1 (elite) · Mookie Betts: 12.4 · Naylor: 9.8 (small lead)

Tier chart

 Aggressive	13.0 +
 Above avg	12.0 - 13.0
 Average	11.0 - 12.0
 Conservative	10.0 - 11.0
 Very small	≤ 10.0

Why it matters

Bigger lead = less distance to cover. But coaches push leads to where the pitcher gets nervous, so big leads can also increase pickoff risk. Notably Naylor has one of the SMALLEST primary leads in MLB (9.8 ft) yet steals 90%+ — his edge is elsewhere.

Source

Savant Basestealing Run Value · `r_primary_lead` (career snapshot — Savant ignores year filter on this

Secondary Lead

What it is

Average distance off first base at the moment of pitch release.

Units	League avg	Elite
feet (ft)	14.5	16.0 +

Real example

Mookie Betts: 18.1 (elite) · Naylor: 13.7

Tier chart

 Elite	16.0 +
 Above avg	15.0 – 16.0
 Average	14.0 – 15.0
 Below avg	13.0 – 14.0
 Poor	≤ 13.0

Why it matters

Primary lead PLUS ground covered while the pitcher is delivering. Big secondary leads compress the steal distance AND give the runner momentum.

Source

Savant · `r_secondary_lead` (career snapshot)

Lead Gain (the 'jerk' metric)

What it is

Distance covered between pitcher's first move and pitch release.

Units	League avg	Elite
feet (ft)	3.5	4.5 +

Real example

Mookie Betts: 5.6 (elite — reads pitchers) · Naylor: 3.9 (above avg)

Tier chart

 Elite	4.5 +
 Above avg	3.8 – 4.5
 Average	3.2 – 3.8
 Below avg	2.6 – 3.2
 Poor	≤ 2.6

Why it matters

The user's 'jerk' metric. Captures TWO things at once: how well the runner READS the pitcher's first move AND how explosively they take their secondary lead. Big lead-gain is a hallmark of smart baserunners (Betts, Chisholm Jr., Frazier, Arraez).

Source

Savant · $r_secondary_lead - r_primary_lead$

v5 / v6 NEW Metrics

Pre-Release Velocity (pre_rel_vel)

What it is

How much ground the runner covers per second between pitcher first move and pitch release.

Units	League avg	Elite
ft / sec	2.7	3.5 +

Real example

Jazz Chisholm Jr. 2025: 4.34 (elite) · Naylor: 2.98 · Soto: 2.39

Tier chart

 Elite	3.5 +
 Above avg	3.0 – 3.5
 Average	2.5 – 3.0
 Below avg	2.0 – 2.5
 Poor	≤ 2.0

Why it matters

Even if a pitcher has a fast delivery, a runner who covers a lot of ground in that small window is still dangerous. We use a league-constant delivery (1.30 s) as the divisor because per-pitch pitcher TTP isn't publicly available — so variation is driven by Lead Gain. The leaderboard separates 'smart' baserunners from pure speedsters.

Source

Lead Gain $\div$ 1.30 s

Post-Release Distance (post_rel_dist)

What it is

How much ground the runner covers during the catcher's pop time.

Units	League avg	Elite
feet (ft)	51.5	58.0 +

Real example

Victor Scott II 2025: 59.5 (elite) · Naylor: 41.5 (last %ile)

Tier chart

 Elite	58.0 +
 Above avg	54.0 – 58.0
 Average	50.0 – 54.0
 Below avg	46.0 – 50.0
 Poor	≤ 46.0

Why it matters

After the pitcher releases the ball, the runner has ~1.95 seconds (the catcher's pop time) to cover as much ground as possible before the throw arrives. Slower runners still accelerating during this window cover LESS than sprint × pop implies — we subtract an acceleration tax for them. This is the speedster-archetype metric: Witt, Carroll, Scott II dominate.

Source

Sprint Speed × catcher pop_2b_sba – acceleration correction

3-2 Count Attempt Share (pct_in_HL)

What it is

Share of a runner's stolen-base attempts that came on a 3-2 (full) count.

Units	League avg	Elite
fraction (0.0 - 1.0)	≈ 0.55	varies

Real example

Naylor 2023: 100% · Naylor 2025: 0% · Most runners: 50-80%

Tier chart

 Heavy 3-2 stealer	0.75 +
 Mostly 3-2	0.50 - 0.75
 Balanced	0.25 - 0.50
 Rarely 3-2	0.05 - 0.25
 Never 3-2	0.00

Why it matters

WHAT THIS REALLY MEASURES — there's a Statcast quirk: when we parse SB attempts from play descriptions, the count attached is the AT-BAT's FINAL count, not the count when the runner actually broke for second. So an SB that happens mid-AT-BAT where the AB ended on 3-2 looks like 'an SB on a 3-2 count' even if it really happened on 0-0. This means pct_in_HL is really 'fraction of this runner's attempts where the AB happened to end on 3-2'. It carries a small signal but mostly reflects how often the at-bat went to full count. v6 plan: fix by pulling MLB Stats API play-by-play.

Source

regex on Statcast `des` field

Real Stolen-Base Performance

What it is

Hard counts: stolen bases and caught stealings in the season.

Units count	League avg varies	Elite 30 SB +
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Real example

Naylor 2025: 30 SB / 2 CS = 93.8% success

Tier chart

 Elite volume	30 + SB
 Above avg	20 - 30 SB
 Average	10 - 20 SB
 Low	< 10 SB (excluded from model)

Why it matters

Ground truth. We require $SB + CS \geq 10$ in a season for the runner-season to qualify (smaller samples are too noisy).

Source

MLB Stats API

Real SB % (shrunk)

What it is

Stolen-base success rate, adjusted to prevent tiny samples from looking 100%.

Units	League avg	Elite
fraction (0-1)	0.78	0.85 +

Real example

Naylor 2025: 0.917 (30 SB, 2 CS)

Tier chart

 Elite	0.88 +
 Above avg	0.80 – 0.88
 Average	0.72 – 0.80
 Below avg	0.65 – 0.72
 Poor	≤ 0.65

Why it matters

Shrinkage: a 1-for-1 runner doesn't look like 100%. Formula: $(SB + 5 \cdot \text{league\%}) \div (SB + CS + 5)$. League-mean prior 'pulls' small samples toward 78%.

Source

computed

Expected SB %

What it is

The SB% you'd expect from this runner based ONLY on their speed.

Units	League avg	Elite
fraction (0-1)	$\approx$ Real SB %	n/a

Real example

Naylor 2025 expected: 0.79 (3rd %ile speed) · actual: 0.92

Tier chart

 **Used as a baseline** compare against Real SB% to get residual

Why it matters

Baseline. Subtracting from Real SB% gives the 'speed-adjusted skill' signal.

Source

2nd-order polynomial fit on Sprint Speed across qualified runners

SB Residual (the KEY signal)

What it is

How much better (or worse) the runner is than their speed predicts.

Units	League avg	Elite
fraction (typically -0.20 to $+0.20$)	0	$+0.10$ or more

Real example

Naylor 2025: $+0.122$ (huge overperformer) · Soto 2025: $+0.128$

Tier chart

	Massive overperformer	$+0.10$ +
	Above expectation	$+0.04$ to $+0.10$
	As expected	-0.04 to $+0.04$
	Below expectation	-0.10 to -0.04
	Far below	≤ -0.10

Why it matters

POSITIVE values mean the runner steals MORE successfully than their raw speed suggests. This is the Naylor / Soto story. We weight it heavily in the SSSI index.

Source

Real SB% - Expected SB%

Battery Context (real per-attempt)

Catcher Pop Time (pop_2b_sba)

What it is

Catcher's time from glove to second base on a steal attempt.

Units	League avg	Elite
seconds	1.95	≤ 1.85

Real example

Patrick Bailey: 1.85 (elite) · league avg: 1.95

Tier chart

Elite arm	≤ 1.85
Above avg	1.86 – 1.92
Average	1.93 – 1.97
Below avg	1.98 – 2.04
Poor	≥ 2.05

Why it matters

REAL per-catcher-per-year data starting 2018. Low pop time = throw arrives at 2B sooner = harder to steal. In Model A this is one of the top 5 features.

Source

Statcast Catcher Poptime leaderboard

Catcher Arm Strength (maxeff_arm_2b_3b_sba)

What it is

Maximum-effort throw velocity to second base.

Units	League avg	Elite
mph	84.0	88.0 +

Real example

Korey Lee 2024: 88.3 mph (elite)

Tier chart

 Elite	88.0 +
 Above avg	85.0 – 88.0
 Average	82.0 – 85.0
 Below avg	79.0 – 82.0
 Poor	≤ 79.0

Why it matters

Component of Pop Time. Some catchers have weak arms but quick transfers; some have rockets but slow exchanges. v5/v6 use both as separate features.

Source

Statcast Catcher Poptime

Catcher Exchange (exchange_2b_3b_sba)

What it is

Time from glove receive to throwing-hand release.

Units	League avg	Elite
seconds	0.64	≤ 0.60

Real example

Patrick Bailey 2024: 0.60

Tier chart

Elite	≤ 0.60
Above avg	0.61 – 0.63
Average	0.64 – 0.66
Below avg	0.67 – 0.69
Poor	≥ 0.70

Why it matters

Fast exchange compensates for a weak arm and vice-versa.

Source

Statcast Catcher Poptime

Pitcher Lead Allowed

What it is

Average lead distance runners take against THIS pitcher (career).

Units	League avg	Elite
feet (ft)	3.5 (lead gain allowed)	How good pitcher is great at suppressing lead

Real example

Best at holding runners → low value; Slow to plate → high value

Tier chart

Suppresses leads	≤ 2.8
Above avg	2.8 – 3.3
Average	3.3 – 3.7
Below avg	3.7 – 4.1
Gives up leads	≥ 4.1

Why it matters

Career-aggregate. When this number is small, the pitcher is good at holding runners. Big = runners feast. Joined to each SB attempt for matchup-level modelling.

Source

Savant Pitcher Running-Game Run Value

Pitcher TTP (Time-to-Plate)

What it is

Seconds from pitcher's first motion to ball-in-catcher's-glove.

Units	League avg	Elite
seconds	1.30	≤ 1.15 (slide-step)

Real example

league constant used in v5/v6 (real per-pitch not published)

Tier chart

 Slide step	≤ 1.15
 Fast	1.16 - 1.25
 Average	1.26 - 1.35
 Slow	1.36 - 1.45
 Very slow	≥ 1.46

Why it matters

Real per-pitch TTP would be the single most useful new data source — but Statcast doesn't publish it. Savant's pitch-tempo CSV is buggy at source (the 'with-runners-on' column equals the 'bases-empty' column for every row). v6 substitutes the league constant 1.30 s.

Source

league constant 1.30 (v6)

Game Situation

Inning

What it is

Which inning the play occurred.

Units	League avg	Elite
integer 1 - 15	—	—

Real example

Late-game (7+) shows different attempt rates than early.

Tier chart

Early	1 - 3
Middle	4 - 6
Late	7 - 9
Extras	10 +

Why it matters

Surprisingly important in Model A — the highest SHAP feature. Late-inning leverage changes manager decisions and runner risk tolerance.

Source

Statcast pitch-level

Outs (outs_when_up)

What it is

Number of outs in the half-inning (0, 1, or 2).

Units	League avg	Elite
0 / 1 / 2	—	—

Real example

Two-out steals are riskier; managers send less often.

Tier chart

0 outs	highest attempt rate
1 out	average
2 outs	lowest — risk of ending inning

Why it matters

Used as a feature; not a huge signal but small reliable effect.

Source

Statcast pitch-level

Balls / Strikes (count)

What it is

Number of balls / strikes BEFORE the current pitch.

Units	League avg	Elite
integers 0-3 / 0-2	—	—

Real example

3-2 (full count) is by far the most common SB-attempt context.

Tier chart

 Full count 3-2	highest by far (1.9% per pitch)
 Two-strike	0.27 - 0.29%
 All other counts	≈ 0% (data quirk — see 3-2 share above)

Why it matters

Feeds into 3-2 Count Attempt Share. See limitation note in that entry — exact count at SB is not in `des` text.

Source

Statcast pitch-level

Composite Scores

SSSI v5 (Slow-Steal Skill Index)

What it is

A weighted composite score that identifies slow-but-effective stealers.

Units	League avg	Elite
z-score units (typically −3 to +3)	0	+2.0 or more

Real example

Naylor 2025: +1.99 · Soto 2025: +1.26 · league mean: 0

Tier chart

Elite slow-steal	+2.0 +
Above avg	+1.0 to +2.0
Average	−0.5 to +1.0
Below avg	−1.5 to −0.5
Poor	≤ −1.5

Why it matters

Weighted average of eight standardized features (see v5 report). Designed to find the Naylor / Soto archetype: elite-performing slow runners. Weights were tuned on 80% of runners with Naylor + Soto HELD OUT to avoid overfitting.

Source

computed in v5/v6 pipeline

How to Read the Model's Weight Table

SB % Boost per Tier (standardised logit coefficient)

What it is

If a runner improves this feature by 1 tier (1 standard deviation), how much does the model's predicted SB success rate go up?

Units	League avg	Elite
percentage points (pp)	0	+5 pp or more = a strongly helpful feature

Real example

Jump Time has -10.6 pp → improving Jump Time by 1 tier raises predicted SB% from 62% to 51% (-10.6 pp). Wait, that's WORSE? No — Jump Time is in seconds, and LOWER is better. So a 1-tier INCREASE (slower) hurts by 10.6 pp. The model shows direction with a sign.

Tier chart

 Strongly helps	+5 pp +
 Helps a little	+1 to +5 pp
 Neutral	-1 to +1 pp
 Hurts a little	-5 to -1 pp
 Hurts a lot	≤ -5 pp

Why it matters

Replaces the old 'coef_z' column. Plain-English version of the same number.

Compute as:

$P_{\text{baseline}} = \text{sigmoid}(\text{intercept})$

$P_{\text{after}} = \text{sigmoid}(\text{intercept} + \text{coef})$

$\text{boost_pp} = (P_{\text{after}} - P_{\text{baseline}}) \times 100$

Source

logistic regression coefficient × baseline probability slope

Odds Multiplier (odds ratio per SD)

What it is

If a runner improves by 1 tier, what happens to their ODDS of success?

Units	League avg		Elite
multiplicative factor	1.0	1.20 +	= feature multiplies odds by 20

Real example

Lead Gain has Odds Multiplier 1.16 → improving by 1 tier multiplies the runner's success-odds by 1.16.

Jump Time has Odds Multiplier 0.63 → 1 tier SLOWER cuts the odds to 63% of baseline.

Tier chart

 Strongly helps	1.20 +
 Helps	1.05 – 1.20
 Neutral	0.95 – 1.05
 Hurts	0.80 – 0.95
 Strongly hurts	≤ 0.80

Why it matters

Replaces the old 'OR/SD' column. Multiply with current odds to get new odds.
Equivalent to $e^{\text{coefficient}}$.

Source

`exp(logistic coefficient)`